

Augustinos D. Saravanos

Postdoctoral Associate
Department of Aeronautics and Astronautics
Massachusetts Institute of Technology (MIT)
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PERSONAL

[🌐 Website](#) | [🔍 Google Scholar](#) | [🌐 LinkedIn](#)

Citizenship: US, Greek

RESEARCH INTERESTS

The Science of Scaling Decisions: At the Intersection of Optimization and AI

Theory: large-scale optimization, (stochastic) optimal control, learning-to-optimize, generative modeling

Applications: multi-agent autonomy, robotics, transportation, operations research

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Machine Learning

2025

Thesis: Distributed Optimization Architectures for Large-Scale Decision Making

Advisor: Prof. Evangelos A. Theodorou

Committee: Profs. Arkadi S. Nemirovski, Yao Xie, Justin Romberg, Efstathios Bakolas

Georgia Institute of Technology

Atlanta, GA

M.Sc. in Aerospace Engineering

2024

University of Patras

Patras, Greece

Diploma in Electrical and Computer Engineering

2019

Thesis: Nonlinear Model Predictive Control for Space Robotic Systems

Co-advisors: Profs. Evangelos Papadopoulos, Nick Koussoulas

Class Rank: 2nd out of 211 (top 1%), GPA: 8.77/10

EXPERIENCE

Massachusetts Institute of Technology (MIT)

Cambridge, MA

Postdoctoral Associate

Aug 2025-present

Supervisor: Prof. Chuchu Fan

Focus: Generative AI for optimization, decision-making and autonomy

Georgia Institute of Technology

Atlanta, GA

Graduate Research Assistant

Aug 2020-Jul 2025

Supervisor: Prof. Evangelos A. Theodorou

Focus: Distributed optimization for large-scale decision-making

BOSCH Center for Artificial Intelligence

Pittsburgh, PA

Machine Learning Research Intern

May-Aug 2023

Supervisor: Dr. Wan-Yi Lin

Focus: Collaborative learning via model alignment

PUBLICATIONS

(* Equal contribution, †Equal advising)

Journal Publications

- [J4] [Second-Order Constrained Dynamic Optimization](#),
Y. Aoyama, O. So, **A.D. Saravanos** and E.A. Theodorou,
International Journal of Robotics Research (IJRR), 2026.
- [J3] [Distributed Covariance Steering via Non-Convex ADMM for Large-Scale Multi-Agent Systems](#),
A.D. Saravanos, I.M. Balci*, A.T. Abdul*, E. Bakolas and E.A. Theodorou,
IEEE Transactions on Automatic Control (TAC), 2026. Submitted.
- [J2] [Scaling Robust Optimization for Swarms: A Distributed Perspective](#),
A.T. Abdul*, **A.D. Saravanos*** and E.A. Theodorou,
IEEE Transactions on Automatic Control (TAC), 2026. Submitted.
- [J1] [Distributed Differential Dynamic Programming Architectures for Large-Scale Multi-Agent Control](#),
A.D. Saravanos, Y. Aoyama, H. Zhu and E.A. Theodorou,
IEEE Transactions on Robotics (T-RO), 2023.

Conference Publications

- [C16] [Asynchronous Distributed Multi-Robot Motion Planning Under Imperfect Communication](#),
A. Tajbakhsh, **A.D. Saravanos**, J. Zhu, E.A. Theodorou, L.T. Biegler and A.M. Johnson,
IEEE International Conference on Robotics and Automation (ICRA), 2027. Submitted.
- [C15] [Language Grounded Bayesian Active Learning in Interactive Robot Planning](#),
H. Li, C. Sobolewski, **A.D. Saravanos**, W. Tan, J. Karigiannis and C. Fan,
Conference on Robot Learning (CoRL), 2026. Spotlight (top 5%).
- [C14] [Learning-to-Optimize via Deep Unfolded Flows](#),
A.D. Saravanos, O. So, H.M. Sabbir Ahmad and C. Fan,
International Conference on Machine Learning (ICML), 2026. Spotlight (top 2%).
- [C13] [Deep FlexQP: Accelerated Nonlinear Programming via Deep Unfolding](#),
A. Oshin, R. Ghosh, **A.D. Saravanos** and E.A. Theodorou,
International Conference on Learning Representations (ICLR), 2026.
- [C12] [Momentum Multi-Marginal Schrödinger Bridge Matching](#),
P. Theodoropoulos, **A.D. Saravanos**, E.A. Theodorou† and G.H. Liu†,
Neural Information Processing Systems (NeurIPS) 2025.
- [C11] [Nonlinear Robust Optimization for Planning and Control](#),
A.T. Abdul, **A.D. Saravanos** and E.A. Theodorou,
IEEE Conference on Decision and Control (CDC), 2025.
- [C10] [Operator Splitting Covariance Steering for Safe Stochastic Nonlinear Control](#),
A. Ratheesh, V. Pacelli, **A.D. Saravanos** and E.A. Theodorou,
IEEE Conference on Decision and Control (CDC), 2025.
- [C9] [Deep Distributed Optimization for Large-Scale Quadratic Programming](#),
A.D. Saravanos, H. Kuperman, A. Oshin, A.T. Abdul, V. Pacelli and E.A. Theodorou,
International Conference on Learning Representations (ICLR), 2025.
- [C8] [Scalable Robust Optimization for Safe Multi-Agent Control Under Deterministic Uncertainty](#),
A.T. Abdul*, **A.D. Saravanos*** and E.A. Theodorou,
American Control Conference (ACC), 2025.

- [C7] [Distributed Model Predictive Covariance Steering](#),
A.D. Saravanos, I.M. Balci, E. Bakolas and E.A. Theodorou,
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.
- [C6] [A Robust Differential Neural ODE Optimizer](#),
P. Theodoropoulos, G.H. Liu, T. Chen, **A.D. Saravanos** and E.A. Theodorou,
International Conference on Learning Representations (ICLR), 2024.
- [C5] [Distributed Hierarchical Distribution Control for Very-Large-Scale Clustered Multi-Agent Systems](#),
A.D. Saravanos, Y. Li and E.A. Theodorou,
Robotics: Science and Systems (RSS), 2023.
- [C4] [Improved Exploration for Safety-Embedded DDP Using Tolerant Barrier States](#),
J.E. Kuperman, H. Almubarak, **A.D. Saravanos** and E.A. Theodorou,
International Conference on Advanced Robotics (ICAR), 2023.
- [C3] [Decentralized Safe Multi-Agent Stochastic Optimal Control using Deep FBSDEs and ADMM](#),
M.A. Pereira*, **A.D. Saravanos***, O. So and E.A. Theodorou,
Robotics: Science and Systems (RSS), 2022.
- [C2] [Receding Horizon Differential Dynamic Programming Under Parametric Uncertainty](#),
Y. Aoyama, **A.D. Saravanos** and E.A. Theodorou,
IEEE Conference on Decision and Control (CDC), 2021.
- [C1] [Distributed Covariance Steering with Consensus ADMM for Stochastic Multi-Agent Systems](#),
A.D. Saravanos, A.G. Tsolovikos, E. Bakolas and E.A. Theodorou,
Robotics: Science and Systems (RSS), 2021.

Workshop Papers / Technical Reports

- [O3] Bayesian Active Learning for Intent Disambiguation in Interactive Robot Planning,
H. Li, C. Sobolewski, **A.D. Saravanos**, W. Tan, J. Karigiannis and C. Fan
IROS Human-Robot Dialogue Workshop, 2026.
- [O2] [Sim2Real on the Robotarium Platform Using Decentralized Multi-Agent Safe Deep FBSDEs](#),
M.A. Pereira*, **A.D. Saravanos*** and E.A. Theodorou,
RSS Workshop on Scaling Robot Learning, 2022.
- [O1] [Sampling-Based Optimization for Multi-Agent Model Predictive Control](#),
Z. Wang, **A.D. Saravanos**, H. Almubarak, O. So and E.A. Theodorou,

Theses

- [T2] Distributed Optimization Architectures for Large-Scale Decision-Making,
A.D. Saravanos
Ph.D. Thesis, Georgia Institute of Technology, 2025.
- [T1] Nonlinear Model Predictive Control for Space Robotic Systems,
A.D. Saravanos
Diploma Thesis, University of Patras, 2019.

PATENTS

- [P3] [Collaborative learning with full model alignment](#)
A.D. Saravanos, F.J. Cabrita Condessa, W.-Y. Lin, Z. Li and M.R. Ganesh
U.S. Patent Application No. 18/371,596.

2023

[P2]	Collaborative learning with full model alignment	2023
	A.D. Saravanos , F.J. Cabrita Condessa, W.-Y. Lin, Z. Li and M.R. Ganesh U.S. Patent Application No. 18/371,594.	
[P1]	Collaborative learning with full model alignment	2023
	A.D. Saravanos , F.J. Cabrita Condessa, W.-Y. Lin, Z. Li and M.R. Ganesh U.S. Patent Application No. 18/371,587.	

GRANT WRITING

[G1]	Safety Guarantees and Stability Verification for End-to-End Autonomous Driving Models	2026
	Hyundai Motor Group (HMG), Project Manager: Dr. Yong-suk Kang Amount: \$70,000 (May-Nov 2026) Role: Lead author (PI: Prof. Chuchu Fan, MIT)	

AWARDS/FELLOWSHIPS

Onassis Foundation Scholar - Doctoral Fellowship	2021-2025
<i>Four-year award for PhD studies financial support</i>	
Gerondelis Foundation - Doctoral Fellowship	2022
<i>One-time award for PhD studies financial support</i>	
Skouras Foundation Scholarship	2019
<i>Top ECE student and in top 10 students of University of Patras for academic year 2018-19</i>	
Valedictorian, Class of 2019, December Graduation Ceremony, ECE, University of Patras	2019
1st & 2nd places, Line Following Robots (Enhanced), Robotex International 2018, Estonia	2018
3rd place, Line Following Robots (Enhanced), Robotex International 2017, Estonia	2017
Bronze medal, European Science Olympiad (EUSO) 2013, Luxembourg	2013
Gold medal, National Science Olympiad 2013, Greece	2013

INVITED TALKS

2026 INFORMS Optimization Society Conference, Computational Optimization Track	Mar 2026
<i>Topic: Deep Learning-Aided Large-Scale Distributed Optimization</i>	
JST ASPIRE Meeting	Mar 2026
<i>Topic: Learning-to-Optimize via Generative Modeling</i>	
Worcester Polytechnic Institute - Robotics Engineering, Guest Lecture	Nov 2025
<i>Topic: Towards Large-Scale Autonomy: Multi-Agent Planning and Control</i>	

MENTORSHIP/SUPERVISION

Graduate Students

Carson Sobolewski, PhD Student at MIT	2025-present
<i>Research topic: Safe multi-agent control for autonomous field inspection robots</i>	
Arshiya Taj Abdul, MS/Junior PhD student at Georgia Tech	2023-2025
<i>Research topic: Distributed robust optimization for multi-agent control under uncertainty</i> <i>Resulted into 1 ACC '25, 1 CDC '25 and 1 TAC (under review) publications.</i>	
Panagiotis Theodoropoulos, Junior PhD student at Georgia Tech	2023-2025
<i>Research topic: Stochastic optimal control and generative AI</i> <i>Resulted into 1 ICLR '24 and 1 NeurIPS '25 publications.</i>	

Akash Ratheesh, Junior PhD student at Georgia Tech <i>Research topic: Operator splitting covariance steering for safe stochastic nonlinear control</i> <i>Resulted into 1 CDC '25 publication.</i>	2023-2025
Taehyun Yoon, Junior PhD student at Georgia Tech <i>Research topic: Distributed trajectory optimization for multi-agent systems</i> <i>Resulted into an upcoming CDC '26 submission.</i>	2024-2025
Joshua Kuperman, MS student at Georgia Tech <i>Research topic: Tolerant barrier states for safe trajectory optimization</i> <i>Resulted into 1 ICAR '23 publication.</i>	2022-2023

Undergraduate Students

Dami Thomas, UROP (Undergraduate Research Opportunities Program) at MIT <i>Research topic: Safe multi-agent control for autonomous field inspection robots</i>	Fall 2025
Hunter Kuperman, BS Student at Georgia Tech <i>Research topic: Learning-to-optimize for quadratic programming</i>	Spring-Fall 2024

TEACHING

AI in Robotics: Learning Algorithms, Design and Safety - MIT Professional Education <i>Co-instructor (with Prof. Chuchu Fan). Co-designed curriculum and delivered a 3-day intensive course for industry professionals on learning-based robotics, control, and safety.</i>	July 2026
Guest Lecturer at Feedback Control Systems 16.30/31, MIT <i>Lecture title: "Advanced Topics: Optimal Control and Lyapunov Analysis"</i>	Dec 2025
Guest Lecturer at Motion Planning RBE 550, Worcester Polytechnic Institute (WPI) <i>Lecture title: "Towards Large-Scale Autonomy: Multi-Agent Planning and Control"</i>	Nov 2025
Volunteer teacher, Robots at MET Program, University of Patras, Greece <i>Taught teams of elementary school students on LEGO robotics</i>	2018-2020
TISP (Teacher In-Service Program) Volunteer, IEEE Student Branch, University of Patras <i>Taught elementary school students basic concepts in engineering</i>	2016-2020
Volunteering course coordinator at "Skagiopouleio" Orphanage, Patras, Greece <i>Organized "Introduction to Coding" course for elementary school students</i>	May 2017

PEER REVIEW SERVICE

Journals: T-RO, TAC, TCNS, RA-L, L-CSS, IJRNC, TSMC.

Conferences: RSS, ICRA, IROS, ICLR, ICML, NeurIPS, L4DC, CDC, ACC, ECC, ICAR, IASEAI.

TECHNICAL SKILLS

Coding Languages:	Python, C/C++, MATLAB, Julia
Packages:	PyTorch, Scikit-learn, NumPy, CVX/CVXPY, MOSEK, Gurobi
IDEs:	Visual Studio, Jupyter Notebook, PyCharm, Spyder

LANGUAGES

English (Fluent), Greek (Native), French (Advanced)

REFERENCES

Prof. Evangelos A. Theodorou

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